# Profile-Based Adaptive XAI for Cognitive Accessibility — Research Approach

Sep 29, 2026 · @Minidu

## Summary (version 3.2, 12 Oct 2026)

We will write an **8-page full paper** for UISE 2027: a requirements-driven way to adapt the explanation of a flood ML prediction to a user-chosen explanation profile. It is built as a React Native prototype and evaluated with a persona-based cognitive walkthrough.

**What changed from version 3.1** (12 Oct; Task 5 replaced by Framework v2)

- Contribution: the paper now leads with run-time fidelity invariants (uncertainty included); capability dependencies are second. User-chosen adaptation and requirement-to-feature tracing are no longer claimed as new, after the related-work check (Madugalla et al. and NeuroAdaptX already do them).
- Alert mode (Layer 3): one short line that keeps all five facts, instead of action only, so the invariants rule has no exception. To confirm with Madam (G124).
- The invariants check runs after every layer, with fallback L4 → L2 → L1; new figure.
- The invariants and capability tables now name the data model fields ( `ews-mobile/model/` ), and mark the two placeholder capabilities.
- Gap statement, NeuroAdaptX differentiation, positioning table and open questions updated to match.

**What changed in version 3.1 (7 Oct 2026)**

- Task 2 now uses Supun's checked evidence: Flood Hub's model gives a distribution, but its public API gives one value per day; the official Google API is cited; GloFAS is added; the Sri Lankan level names are confirmed; the Mosavi claim is corrected and Bentivoglio et al. (2022) added.
- The scenario is fixed: Hanwella gauge, Kelani Ganga; ABOVE_NORMAL with a 0.29 chance of reaching the Minor Flood Level, up from NO_FLOODING at 0.10 yesterday.
- Task 2b: the expected table is replaced by the real SHAP, LIME and DiCE results (first-pass ratings; second rating due 12 Oct).
- Madugalla et al. checked against the full text; the positioning table is updated. The CAP 1.2 and NeuroAdaptX checks of 6 Oct are carried in.
- The worked example, the worked trace and the capability table use the real scenario numbers.

**What changed from version 2**

- Task 1 triage is now Minindu's manual evaluation of all 128 guidelines, not a first-pass proposal.
- The guideline tables moved to a separate evidence catalogue, with how each guideline is adapted in our flood EWS app and the published evidence for it.
- 19 guidelines from the manual review of the catalogue were added to the 31 from the first triage (50 in total); none was removed.

**What changed from version 1**

- **Paper type:** 5-page vision paper → 8-page full paper, because we now build and evaluate a prototype.
- **Prediction types are grounded:** chosen from the literature and from deployed AI flood systems (Google Flood Hub), not picked by us.
- **New motivation study:** we run SHAP, LIME and a counterfactual on a synthetic flood model and show which of our guidelines their output fails.
- **Profile questions are grounded:** built from the cognitive domains in standard screening tools and from validated self-report scales, not from the tests themselves (see Task 3 for why).
- **Three profiles are justified** from existing apps (Competitive Analysis 04) and prior XAI work.
- **Prototype:** two functions (set up / select profile; view prediction), wireframes first, then AI-assisted build on our existing React Native EWS app, same colours as v2.
- **Evaluation:** cognitive walkthrough with personas built from the three profiles. Evaluators record problems in plain words; we map them to guidelines afterwards.

**Unchanged:** keep the uncertainty ("Possible", not deleted), profiles by need not diagnosis, fidelity invariants as the core SE idea, Brief 2 (Vedda) stays a separate paper.

The schedule and owners are in the separate plan doc, "Adaptive XAI Paper — Feasible Plan to 13 Nov".

## Research question and scope

**Central question (Brief 1):** How can cognitive accessibility requirements be identified, represented as XAI preference profiles, and used to adapt the explanation of disaster-related ML predictions?

**Sub-questions:**

1. Which existing cognitive accessibility requirements are relevant to XAI?
2. Which disaster ML prediction types create different explanation requirements?
3. How can XAI profiles determine explanation content, detail and interaction?

**Motivating example.** A standard output reads: "In two days there is a 30% chance your area will flood due to current weather." For someone with a cognitive impairment this may be hard to process: a percentage, a relative time, an implied location and no action. The adapted version keeps every fact but changes order, form and depth. A worked example is under Task 3.

### In scope

- People with cognitive impairments (dementia, MCI, autism, ADHD and others), described by explanation *needs*, not diagnosis.
- One or two flood prediction types, with synthetic values clearly labelled.
- A literature-grounded requirements synthesis, profile specification, Figma demonstration and framework.

### Out of scope for this paper

- Changing the ML model or its prediction.
- Claims of validation with users (no study has been run).
- Indigenous communities such as the Vedda. Their needs (source, provenance, time period, community governance) are a different problem, covered by Research Brief 2 as a separate paper.

## Positioning and the gap

XAI for people with cognitive disabilities already exists, so our novelty has to be the combination of safety-critical predictions, a requirements process and fidelity checking.

| Work | What it does | What it leaves open for us |
|---|---|---|
| NeuroAdaptX (Bunde, DESRIST 2026) | Adapts explanation structure, modality and density to self-reported condition profiles (ADHD, autism, dyslexia); derives 10 design requirements from cognitive theory; randomised online vignette experiment, N=216, in a digital-wellbeing task; checks informational equivalence with a pre-deployment checklist | Closest competitor. Low-stakes domain, profiles by condition label, requirements from theory not an accessibility catalogue (traced to features, but not to data capabilities), presentation only, equivalence checked once by hand before deployment on fixed templates, without uncertainty (p. 215). Must be cited and contrasted. |

| Work | What it does | What it leaves open for us |
|---|---|---|
| Tielman et al., "Explainable AI for all" (Technology in Society, 2024) | Roadmap for inclusive XAI for people with cognitive disabilities; recommends personalised, adaptive explanations | No disaster domain, no requirements specification or profile structure |
| ProfileXAI (Corrales et al., 2025) | SHAP/LIME/Anchor + LLM explanations for three user types (ML engineer, domain expert, lay user) | Profiles by expertise, not accessibility; healthcare |
| Accessible XAI survey (arXiv 2407.17484) | Surveys accessibility of XAI, mostly visual impairment | Cognitive accessibility under-covered |
| Madugalla et al. (2025), human-centred RE for critical systems (arXiv:2511.12856) | Four-step RE process for disaster EWS: 62 guidelines → 67 requirements → adaptive prototype → validation by 6 interviews and persona walkthroughs (8 evaluators, 4 personas) | States that cognitive impairments were "outside the scope of this study"; the prototype shows a four-day fire danger forecast but never explains it; no XAI. Users already choose their own settings (p. 11), so user choice is not our novelty. Our walkthrough extends their method. Preprint: check for a published version. |
| Our earlier paper | RE process + prototype for cognitively accessible EWS | Warns and guides evacuation, but never explains a prediction |

**Gap statement for the paper:** no work turns cognitive accessibility requirements into explanation requirements for safety-critical ML predictions, delivers them through need-based profiles, and checks at run time that every adapted explanation, including AI wording, keeps five facts of the prediction, its uncertainty among them.

**Differentiation from NeuroAdaptX in one line:** they show that presentation-level adaptation, checked for informational equivalence once before deployment on fixed templates (Bunde 2026, p. 215), improves comprehension in a low-stakes wellbeing task; we move this to safety-critical warnings, with profiles built from explanation needs rather than condition labels, and check five fidelity invariants, including uncertainty, at run time on every adapted explanation. Details: `related-work-notes.md`.

# What our earlier work already shows

Across our paper, interim report, four competitive analyses, 128-guideline catalog and prototype, nothing explains a prediction. This is the new gap, and our own material is the evidence for it.

| Source | Relevant finding | Use in the new paper |
|---|---|---|
| Final paper ("Designing Inclusive Disaster EWS…") | 128-item catalog from 969 guidelines / 49 sources; rule-based profiles; AI only for optional plain-language rewriting | Starting point for Task 1; the prior-work paragraph |
| Competitive Analysis 01 | 5 production EWS apps (FEMA, MyShake, Google Public Alerts, Red Cross, Disaster Alert): none cognitively accessible; plain-language threat descriptions partial or absent | Evidence that mainstream EWS output is not written for this population |
| Competitive Analysis 02 | 13 apps split into predictive EWS vs cognitive-accessibility apps; none does both | Extend the argument: even the predictive apps never explain their predictions |
| Competitive Analysis 03 | Four-layer architecture (baseline → profile rules → alert mode → optional AI); "accessibility illusion" risk of LLM simplification; only 6% of AI-in-EWS work targets dissemination | The architecture the new framework plugs into; motivation for fidelity invariants |
| Competitive Analysis 04 | UI techniques across 7 apps; MyShake shows magnitude colours, Disaster Alert map symbols, no "why" or "how sure" anywhere | Evidence for the explanation gap |
| 128-guideline catalog | Covers how to present (plain language, numbers in several forms, progressive disclosure), not what to explain | Task 1 triage (next section) |

| Source | Relevant finding | Use in the new paper |
|---|---|---|
| Prototype v2 (Figma) | Onboarding picks a condition (Dementia, Autism, MCI, ADHD, Schizophrenia); alert states Safe / Warning / Danger; danger screen says "Danger! Flood coming!" | Base for Task 4; needs a forecast state and an explanation-profile screen |

The Figma file itself could not be opened by automated tools. Prototype observations come from Figure 3.2 of the interim report and Figure 2 of the final paper.

## Task 1: Which requirements matter for XAI

Minindu manually evaluated all 128 guidelines and found 50 that affect how a prediction is explained: 31 from the first triage (10 XAI-related, 21 both) and 19 more from a second manual review of the colour-coded catalogue. The rest are general UI.

The full list is in a separate document, the XAI guideline evidence catalogue (also exported as a PDF). For each guideline it gives the implication for the explanation, how to adapt it in our EWS app, and published evidence: 38 have direct evidence, 9 indirect, and 3 have none found and stay as our inference.

Classes follow Brief 1: **A** = general UI, **B** = XAI-related (changes what is explained or how it is understood), **C** = both. None of the 49 source papers discusses AI explanations, so every B and C implication is *inferred by us*. The paper must say this.

**Why check the evidence again.** The 128 guidelines are already sourced, but their sources answer a different question: does this help people with cognitive impairments use an interface or follow a warning? We apply them to a new object, the explanation of an ML prediction, and that transfer is our inference. The evidence catalogue tests the transfer: does any study show the property changes how people understand an AI explanation, a probability or a warning? Direct evidence supports the transfer, indirect evidence supports it only partly, and "none found" marks it as inference.

### What the catalog does not cover

The catalog explains *how* to present information, never *what* to explain. A targeted literature search is needed for:

- **Reasons and evidence:** why the risk is high, which observed conditions support it. Feature importance must not be presented as cause.
- **Uncertainty for low numeracy:** icon arrays, natural frequencies, verbal probability words and how they are misread.

- **Forecast change over time:** explaining upgrades and downgrades.
- **Source and provenance:** who issued the forecast and from what data.
- **False alarms and trust:** explaining a flood that did not happen so the next warning is still heeded.
- **Impact-based framing:** what the flood will *do* to the user, not what the weather will *be* (WMO impact-based forecasting).
- **Fidelity of simplified text:** checking an Easy Read or LLM version keeps the facts.

## Task 2: Prediction types, grounded in literature and deployed AI

We choose the two output types that the leading deployed AI flood system actually produces: a **severity class** and a **multi-day river forecast with uncertainty**. That makes the choice defensible rather than ours.

### Evidence from deployed AI systems

| System | Model | What it outputs | Type it supports |
|---|---|---|---|
| Google Flood Hub model (Nearing et al., Nature 2024) | LSTM encoder–decoder trained on 5,680 gauges; inputs ECMWF weather, satellite rain, basin attributes | Daily streamflow up to 7 days as a probability distribution (inside the model); 5-day reliability similar to GloFAS same-day nowcasts; free forecasts in over 80 countries | Time-series forecast with uncertainty |
| Flood Hub API (Google Flood Forecasting API) | Same | Gauge severity NO_FLOODING / ABOVE_NORMAL / SEVERE / EXTREME / UNKNOWN; trend; forecasts as one value per interval (no range); inundation maps HIGH / MEDIUM / LOW, meaning probability on some maps and depth on others; Sri Lanka covered | Risk classification; spatial prediction |

| System | Model | What it outputs | Type it supports |
|---|---|---|---|
| GloFAS (Copernicus EMS) | LISFLOOD hydrological model driven by a 51-member weather ensemble (not ML) | Up to 15 days; probability of exceeding 2-, 5- and 20-year return-period flows | Public probabilistic forecast |
| MyShake (Kong et al., 2016; Competitive Analysis 03) | Neural network on phone accelerometers | Earthquake vs everyday motion, then alert (alert pathway to check against CA03) | Event classification |
| Sri Lanka Irrigation Department river bulletin (8 Aug 2026) | Threshold rules (not ML) | Each gauge against Alert Level, Minor Flood Level and Major Flood Level, with Rising or Falling | The local warning vocabulary users already hear (confirmed) |

## Evidence from literature

- Nearing et al. (2024): the main citation. The model produces a probability distribution per day, but the public API exposes one value per interval. Say "the model produces", not "Flood Hub shows".
- Mosavi, Ozturk & Chau (2018), *Water* 10(11), 1536: reviews 180 papers; continuous targets (water level, streamflow) at short lead times dominate and "are used as warning systems". It does **not** support "classification is dominant" (checked by Supun, 6 Oct).
- Bentivoglio et al. (2022), *Hydrology and Earth System Sciences* 26: 58 deep-learning flood mapping papers use classification and regression, and "all reviewed models and their outputs are deterministic". Cite this for classification.
- Recent XAI flood studies apply SHAP to flood susceptibility models, e.g. Choubin et al. (2025), *Results in Engineering* (volume to confirm), which shows feature attribution is the standard explanation in this domain.

## Choice and rationale

| Prediction type | Chosen? | Why |
|---|---|---|
| Risk classification | Yes | Flood Hub's severity classes; maps directly to CAP severity and local warning levels |

| Prediction type | Chosen? | Why |
|---|---|---|
| Time-series forecast | Yes | Flood Hub's core output; carries uncertainty and change between forecasts |
| Spatial prediction | Future work | Needs maps, which are hard for this population; adds a third explanation problem |
| Event likelihood (landslide) | Out of scope | Different hazard; keep one hazard for the paper |

**Scenario (synthetic, labelled in the app):** a Flood Hub–style forecast for the Hanwella gauge on the Kelani Ganga. Today ABOVE_NORMAL, with a 0.29 chance (about 30%) of reaching the Minor Flood Level (7.5 m) on Thursday; yesterday NO_FLOODING at 0.10. Treating the Minor Flood Level as Flood Hub's danger threshold (SEVERE) is our inference. Thresholds are modelled on the Irrigation Department bulletin of 8 Aug 2026.

## Task 2b: What standard XAI gives vs what our guidelines need

This study motivates the framework: we run the standard XAI methods on the same flood prediction and show which explanation requirements their output does not meet.

**Method**

1. Synthetic data, 8,000 rows: 72-hour rainfall, upstream level, soil moisture and the previous day's level at the gauge. Label: the gauge reaches the minor flood level two days later. A gradient boosting classifier, constrained so the probability rises with each input; ROC AUC 0.85 on held-out synthetic rows.
2. Explain one held-out prediction (p = 0.29, chosen by a rule fixed in advance) with **SHAP** (Lundberg & Lee, 2017), **LIME** (Ribeiro et al., 2016) and a **counterfactual** from DiCE (Mothilal et al., 2020; Wachter et al., 2017). Anchors dropped.
3. Record each method's raw output exactly as produced, as a paper figure.
4. Score each output against seven guideline needs plus a hazard row, so all five invariants are covered. Two authors rate independently and report agreement.

**Result** (first-pass ratings by Supun, 6 Oct; second rating due 12 Oct):

| Need (guideline) | SHAP | LIME | Counterfactual (DiCE) | Our adapted explanation |
|---|---|---|---|---|
| Bottom line first (G21) | No: attribution bars; prediction only as f(x) | Partly: names the class | Partly: leads with the outcome code | "Flooding is possible here on Thursday." |
| Likelihood without numeracy (G26, G119) | No: a decimal | No: decimals | No: class only, the 0.29 is lost | "Possible" + 3 of 10 picture |
| Place and time window (G73) | No | No | No | "Here, Thursday" |
| Consequence and action (G50) | No | No | No: changes the model's answer, not the user's action | "Get your go-bag ready" |
| Familiar references (G38) | No: variable names and units | No: bin edges from training data | Partly: compares with current values, raw units | River vs landmark |
| What changed (G89) | No: compares with the data average | No | Partly: hypothetical contrast, not yesterday | "Yesterday it was 1 in 10" |
| Why, in plain words (G25) | Partly: ranked inputs as names and decimals | Partly: rules, but its surrogate misfits (0.58 vs 0.29) | Partly: "if X were Y", as a table | Reasons from observations |
| Hazard named (G11, G120) | To rate | To rate | To rate | Flood symbol + "Flooding" |

**Findings.** SHAP ranked yesterday's river level first and LIME the upstream level, so "the main reason" depends on the method chosen. LIME's local surrogate predicted 0.58 where the model gave 0.29 (to be rechecked over 5 seeds). DiCE's nearest flood case needs about 15 mm more rain and the river 1 m higher upstream (0.57). This is one synthetic case and one toy model: it shows output formats, not evidence about real floods.

The point for the paper: standard XAI answers "which input features moved the model", for developers. Our guidelines need "what does this mean for me, how sure, and what do I do". That gap is why a requirements-driven adaptation layer is needed.

## Task 3: XAI profiles, grounded in literature

The standard cognitive tests cannot be used as the profile itself. Their cognitive domains, and validated self-report scales, give each profile question its literature backing.

### Can we use the Alzheimer's Association screening tools?

The Alzheimer's Association list covers patient tests (Mini-Cog, GPCOG part 1), informant tools (AD8, GPCOG part 2, Short IQCODE) and FDA-cleared digital tests. Using them directly fails on five counts:

| Problem | Evidence |
|---|---|
| They are diagnostic screens given by clinicians, not app setup | Mini-Cog: "healthcare provider after brief training"; results decide whether a full diagnosis is needed (Alzheimer's Association) |
| Not valid for people with intellectual disability | General dementia tests cannot separate dementia from pre-existing disability; ID-specific tools such as DSQIID are needed (Zeilinger et al., 2022) |
| Biased by education and language | MMSE should not be used clinically without about grade-8 education and English fluency (MMSE overview) |
| Licensing | MMSE is copyrighted and sold through PAR (same source) |
| A score measures deficit, not explanation preference | Brief 1 forbids assigning profiles from diagnosis; catalog G84 forbids labelling by condition |

**What we use instead:**

1. **The cognitive domains these tests measure** (memory, attention, executive function, orientation to time and place, language, numeracy). Each domain gives one profile question, so every question traces to a recognised domain.

2. **Validated self-report preference scales** that ask about preferences without testing ability. The model is the Subjective Numeracy Scale (Fagerlin et al., 2007), "measuring numeracy without a math test", which includes items on preferring numbers vs words.
3. **The informant idea** (AD8, IQCODE): a carer can answer or help, which matches G107.
4. **Function, not diagnosis:** WHO's ICF describes disability by functioning, which supports need-based profiles (**Supun to add the citation**).

## The question template

Every question follows one pattern, so one well-made question becomes the model for the rest:

> **Domain → explanation decision it controls → Easy Read question with picture options → "Not sure, choose for me" default → "Ask my helper" option**

**Worked question (numeracy → likelihood form).** Grounded in the Subjective Numeracy Scale's preference items, G26 and G119.

- Question: "When we tell you how likely a flood is, what helps you most?"
- Option A: words ("Possible", "Likely") — picture of the word card
- Option B: words and a picture (3 of 10 houses)
- Option C: words, picture and the number (30%)
- Default if unsure: B. The user can change it later.

## All questions from the template

Each question now traces to the guidelines it serves and to the published evidence behind them, from the evidence catalogue. **D** = direct evidence, **I** = indirect evidence; *new* = added in the second manual review.

| Domain (screening-test source) | Explanation decision | Question (draft) | Guidelines (evidence) | Literature backing |
|---|---|---|---|---|
| Numeracy (SNS) | Likelihood form | When we tell you how likely a flood is, what helps you most? | G26 (D), G119 (D), G81 (D, *new*) | Icon arrays help low-numeracy and older adults (Galesic et al., 2009); probability words are read inconsistently (Budescu et al., 2009); "30% chance" is read in contradictory ways (Gigerenzer et al., 2005) |
| Attention (Mini-Cog, MoCA attention items) | Detail on first view | Do you want only the main message first, or everything at once? | G41 (D), G21 (D), G11 (D), G18 (D, *new*) | Less, focused information improves comprehension, most for low numeracy (Peters et al., 2007); complex explanations are harder to interpret (Lage et al., 2019) |
| Executive function / planning (Mini-Cog clock drawing) | Action detail | Do you want us to tell you exactly what to do, step by step? | G4 (D), G50 (D), G22 (D, *new*) | Explanations explored step by step were understood better (Mindlin et al., 2024); protective action guidance improves understanding (Sutton et al., 2021); list instructions are recalled better (Morrow et al., 1998) |

| Domain (screening-test source) | Explanation decision | Question (draft) | Guidelines (evidence) | Literature backing |
|---|---|---|---|---|
| Memory (Mini-Cog recall; AD8 "repeats questions") | Reminders and change alerts | Should we remind you of the warning and tell you if it changes? | G109 (I), G89 (D), G54 (D), G27 (I, *new*) | Earlier forecasts anchor judgments of new ones (Herdener et al., 2018); repetition helps people with dementia learn (Creighton et al., 2013); too many alerts get ignored (Ha et al., 2026) |
| Orientation to time and place (MMSE orientation) | Time and place wording | How should we say when? "Thursday morning" or "in 2 days"? | G47 (D), G73 (D), G114 (D) | Orientation and calculation are among the most impaired items in MCI and dementia (Liu et al., 2019); location in a warning changes risk judgments (Klockow-McClain et al., 2020); time indications improve understanding (Dallo et al., 2022) |
| Language | Modality and reading level | Do you like to read, listen, or see pictures? | G128 (D), G67 (D), G14 (D, *new*), G15 (I, *new*), G17 (I, *new*) | Text + visual explanations help lay users most (Szymanski et al., 2021); matching the preferred format improves understanding (Tait et al., 2012); jargon impairs processing (Bullock et al., 2019) |

| Domain (screening-test source) | Explanation decision | Question (draft) | Guidelines (evidence) | Literature backing |
|---|---|---|---|---|
| Reasoning / curiosity | Explanation goal | Do you want to know why the app thinks a flood may come? | Brief 1; G25 (D) | Adaptive XAI is the key to inclusive XAI (Tielman et al., 2024); lay users cannot be expected to read a causal chain (Miller, 2019); impact information improves warning understanding (Weyrich et al., 2018) |

## Guidelines without published backing (3)

Three of the 50 XAI-relevant guidelines were marked as possibly relevant, but the evidence search found no source linking them to understanding an explanation, forecast or warning. They stay in the requirements as **our inference**, and the paper must label them that way.

| ID | Guideline | Why it was marked | Where this document relies on it | How we treat it |
|---|---|---|---|---|
| G20 | Include symbols and letters necessary to decipher | Sinhala and Tamil explanations must render correctly, including numerals, % and dates | Supports G121 (local-language Easy Read) | An engineering requirement: test that every explanation string renders in all three languages. Closest source is context only: CAP alerts lost integrity on small handsets in Sri Lankan trials (Waidyanatha et al., 2007), with no comprehension measure. |
| G42 | Info to prepare for a task | State the expected timing and when the next forecast update arrives | Time-window invariant (with G73); "Next update: tomorrow at 6 am" in the "How sure, and what changed?" profile | The invariant stays grounded through G73, which has direct evidence. Next step: a targeted search on lead-time and update-time communication. |
| G65 | Focus Not Obscured (WCAG 2.4.11) | "Why?" and "How sure?" must never be hidden under a banner | Layout rule only | Treat as general UI (class A) in the paper; keep it in the prototype as a WCAG check. |

## Three profiles, and why these three

The profiles are bundles of default answers that the user can still change. Each bundle mirrors a pattern seen in existing apps or prior XAI work.

| Profile | Answers it bundles | Why this profile exists |
|---|---|---|
| "Just tell me what to do" | Main message first; step-by-step actions; words only; listen | One-action-per-screen apps (Emergency Chat, MapHabit, GoTDAH) and calm "overwhelm mode" in Unmasked (Competitive Analysis 04); G124 |
| "Show me why" | Reasons on; words + picture; landmark comparisons | Photo/visual-step design in MapHabit and Dementia Emergency (CA04); the "why" goal in Tielman et al. 2024 and NeuroAdaptX |
| "How sure, and what changed?" | Words + picture + number; change alerts on | Countdown/time externalisation in GoTDAH, data visualisation in Unmasked (CA04); uncertainty needs in the XAI literature |

In the paper these profiles are presented as **hypothetical**, derived from literature and apps, not from users.

## Worked example: one 30% prediction, three explanations

Same synthetic prediction everywhere: ABOVE_NORMAL, a 0.29 (about 30%) chance of flooding in this area on Thursday, up from 0.10 yesterday.

**Standard output:** "In two days there is a 30% chance your area will be flooded due to the current weather."

**"Just tell me what to do"**

1. "Flooding is possible here on Thursday. It is not certain."
2. "Get your go-bag ready today."
3. Button: "Why?" (opens the reasons)

**"Show me why"**

1. "Flooding is possible here on Thursday."
2. "Why: a lot of rain has fallen upstream, and the river is rising."
3. River-level picture against the bridge marker (synthetic).
4. Button: "How sure is this?"

**"How sure, and what changed?"**

1. "Flooding is possible here on Thursday. 3 in 10 chance." (icon array: 3 of 10 houses filled)

2. "Yesterday it was 1 in 10. It went up because of new rain."
3. "Next update: tomorrow at 6 am."
4. Exact figure on tap: 29%.

All three keep the hazard, the place, the time, the likelihood and the action. Only order, form and depth change.

**Why "Possible":** CAP 1.2 (OASIS, 2010, §3.2.2) defines Possible as "Possible but not likely (p <= ~50%)" and Likely as "Likely (p > ~50%)", so 30% → "Possible" is a standard mapping. Verified against the specification on 6 Oct 2026. Because Unlikely means p ~ 0, every value up to 50% is "Possible", including yesterday's 10%, which is why the picture or number stays available. Our cut between Unlikely and Possible (5%) is our own choice and is labelled so.

## Task 4: Prototype (React Native) from wireframes

We extend our existing React Native EWS app with two functions. Every screen is wireframed first, because AI-generated builds from text prompts alone came out generic in our earlier work.

**Function 1: Set up and select an explanation profile**

- First-time setup: the template questions, one per screen, with picture options and "Not sure, choose for me".
- Pick one of the three ready profiles instead of answering questions.
- Change the profile later from settings; carer-assisted setup (optional).

**Function 2: View a prediction and its explanation**

- A new "Forecast" state before Warning, holding "Flooding is possible here on Thursday".
- The same synthetic prediction rendered per profile; "Why?" and "How sure?" on request.
- When it becomes a live alert, every profile collapses to action only (G124).
- A "Synthetic data" label on every forecast screen.

**Build method**

1. Wireframe every screen (hand sketches or a tool such as Excalidraw or Figma).
2. Write a short spec per screen: purpose, components, text, which profile attribute it reads.
3. Prompt the AI with the wireframe image + spec + the v2 colour tokens and component list; one screen per prompt.

4. Review each generated screen against its wireframe before merging.

## Task 4b: Cognitive walkthrough

We evaluate the prototype with a persona-based cognitive walkthrough, adapting the method from our earlier paper, Wharton et al. (1994), IxDF guidance, Madugalla et al. (2025) and the persona-based walkthrough in Huynh et al. (COMPSAC 2021).

**Tasks** (defined from the two functions, before the walkthrough):

| ID | Task | Function |
|---|---|---|
| T1 | Set up an explanation profile for the first time | 1 |
| T2 | Switch to a different ready-made profile | 1 |
| T3 | Read a new forecast and know what, where, when, how likely, and what to do | 2 |
| T4 | Ask for more explanation ("Why?" or "How sure?") and return | 2 |
| T5 | Notice that the forecast has changed since yesterday | 2 |

**Personas:** one per profile, built like our earlier evidence-record personas (affected domains, documented behaviours, urgency modifier, sources), plus an optional carer persona.

**Questions per step:** the four standard questions plus our fifth (detect and recover from an error). For T3 we add a comprehension check: can the persona state the five invariants after reading?

**Evaluator sheet (simpler than last time):**

| Column | What the evaluator writes |
|---|---|
| Task, step, persona | Pre-filled |
| Q1–Q5 | Yes / No |
| What went wrong | Plain words |
| Severity | 0–4 |
| Screenshot | Required for severity 3–4 |

**No guideline column.** Evaluators only describe problems; the team maps each defect to a guideline afterwards, in a separate pass.

**Bias control:** three members walk independently; the person who built a screen does not lead it; severities are rated before discussion, and agreement is reported.

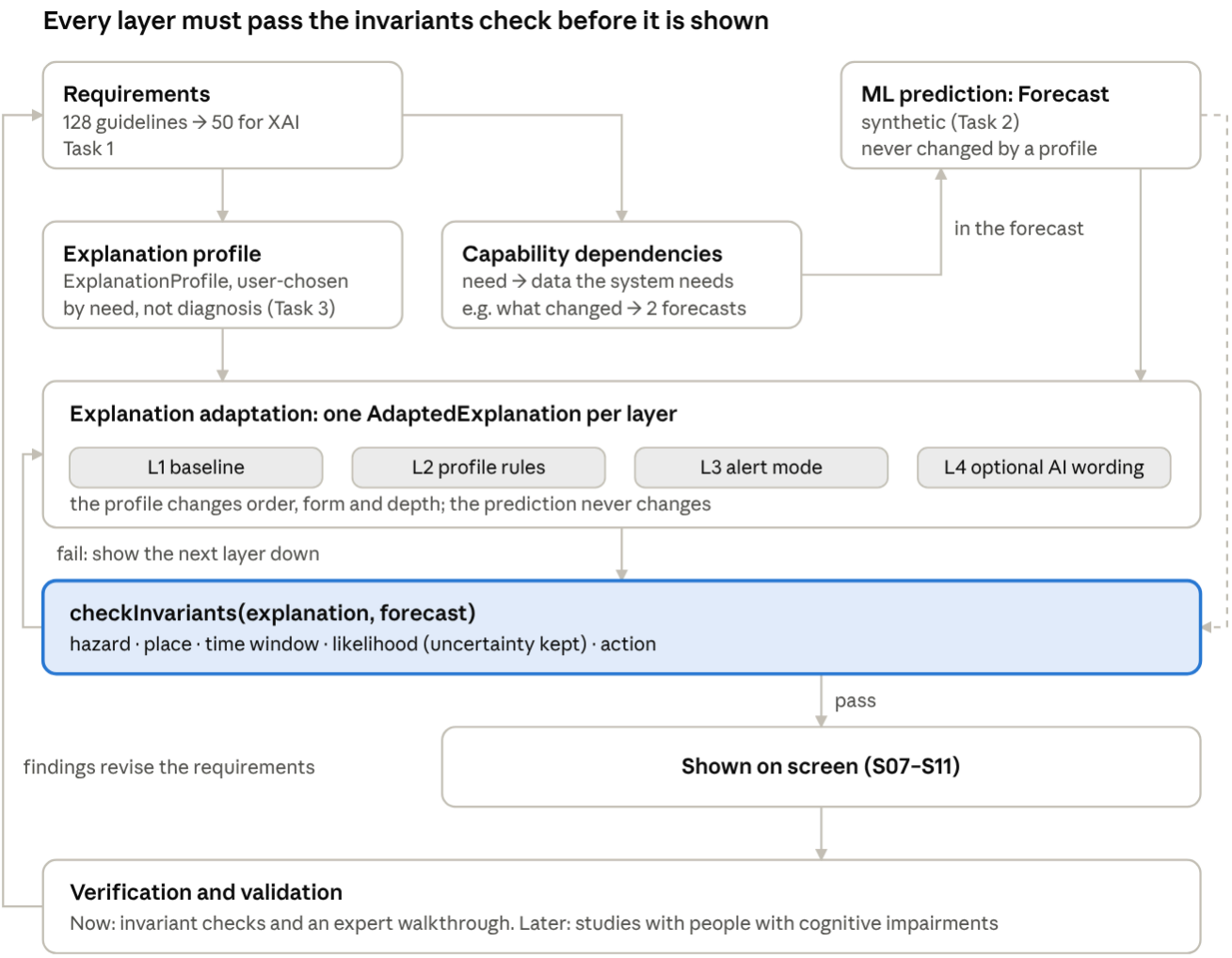
## Task 5: The framework

The framework turns cognitive accessibility requirements into explanation behaviour for an ML flood prediction, and checks every adapted explanation against five fidelity invariants at run time, before anyone sees it. The prediction never changes; only its explanation does. (Version 3.2: replaced by Framework v2, 11–12 Oct.)

The paper claims two things as ours:

1. **Run-time fidelity invariants.** Hazard, place, time window, likelihood and action are checked on every adapted explanation, including AI wording, against the forecast it explains. Uncertainty is one of the invariants, so no profile can drop it. NeuroAdaptX checks equivalence once, before deployment, on fixed templates, with four units that do not include uncertainty (Bunde 2026, p. 215).
2. **Requirement → capability dependencies.** Each explanation need creates data and functions the system must have, traced to fields in the data model.

User-chosen adaptation (Madugalla et al.) and requirement-to-feature tracing (NeuroAdaptX) are part of the framework but cited as prior work, not claimed as new.




Framework v2 · requirements and prediction meet in the layers; every layer is checked

Requirements flow down to a profile and to capabilities the forecast must provide. The profile and the forecast meet in the layers, and each layer's output is compared with the forecast (dashed line) before it is shown. Verification findings feed back into the requirements.

## Fidelity invariants

No profile, simplification, alert mode or AI rewording may remove or change these five facts. Each is a slot in `AdaptedExplanation`; `checkInvariants()` compares it with the `Forecast`.

| Invariant | Example (Hanwella) | Grounding | How it is checked |
|---|---|---|---|
| Hazard | Flooding | G11, G119 | Slot present; value equals `forecast.hazard` |

| Invariant | Example (Hanwella) | Grounding | How it is checked |
|---|---|---|---|
| Place | "here" (Hanwella gauge, Kelani Ganga) | G73 | Slot present; gauge id equals `forecast.place.gaugeId` |
| Time window | on Thursday | G73, G42 | Slot present; start and end equal `forecast.window` |
| Likelihood (uncertainty kept) | "Possible", plus 3 of 10 houses or 29% per profile | G26, G119 | Slot present; CAP word equals `forecast.likelihood.capCertainty`; probability equals the forecast's. For AI wording, the CAP word must also appear in the text |
| Action | Get your go-bag ready today | G11, G50 | Slot present; steps equal `forecast.action.steps` |

Result: pass, or fail with two lists, *missing* (slot absent or text empty) and *wrong* (value differs from the forecast). Any fail shows the next layer down instead.

## Requirement → capability dependencies

An explanation need is not only a UI choice: it requires data and functions in the system. The last columns show where each capability lives in the data model, so every row is checkable in code.

| Explanation need | Guidelines | Capability the system needs | Where it lives | Status |
|---|---|---|---|---|
| What changed since last time | G89, G54 | Stored previous forecasts and a comparison | `ForecastFile` (two forecasts per gauge); `compareForecasts()` | In the model |
| Why is the risk high | G25, G14 | Model exposes supporting observations, never feature importance as cause | `Forecast.supportingObservations` | In the model; values from Supun |
| How sure | G26, G119, G81 | Probability or range from the model | `Forecast.likelihood.probability`, `Forecast.peakLevel` | In the model |

| Explanation need | Guidelines | Capability the system needs | Where it lives | Status |
|---|---|---|---|---|
| Level against a landmark | G38, G76 | Local reference levels for each area | `Forecast.landmark` | Placeholder; no real levels exist |
| When is the next update | G42 | Forecast schedule metadata | `Forecast.nextUpdateAt` | In the model |
| What to do | G11, G50, G22 | Action steps per likelihood level, from a cited source | `Forecast.action` (`source`, `placeholder`) | "Possible" only; other levels placeholder |
| Carer explanation | G107, G80 | Consent record and a second profile | `ExplanationProfile.carerConsent`, `answeredBy` | Consent in the model; second profile not yet |

## The four layers

Every layer produces an `AdaptedExplanation`, and every layer's output must pass `checkInvariants()`; a failed layer falls back to the one below it.

| Layer | What it does | If the check fails |
|---|---|---|
| L1 Baseline | One fixed accessible explanation containing all five invariants | Nothing below it: a fail is a build error, caught by tests |
| L2 Profile rules | Deterministic rules choose content, order, form and depth from the profile; auditable | Show L1 |
| L3 Alert mode | When an alert is live, every profile collapses to one short line with all five facts, e.g. "Flooding possible here on Thursday. Get your go-bag ready today." (G124; to confirm with Madam) | Show L1 |
| L4 Optional AI wording | A language model rewords the L2 text | Show L2 |

## Worked trace ("How sure, and what changed?" profile)

1. **Requirement** G26: provide alternatives for numerical concepts.
2. **Explanation need:** understand likelihood without relying on numeracy.

3. **Profile attribute:** `likelihoodForm = 'wordPictureNumber'` (preset `howSure`).
4. **Prediction:** today p = 0.29, `ABOVE_NORMAL`; yesterday p = 0.10, `NO_FLOODING` (synthetic). `capCertaintyFor()` gives "Possible" for both.
5. **Capability:** `compareForecasts()` gives class changed, direction up, 1 → 3 of 10 houses.
6. **L2 rule:** CAP word "Possible" + icon array 3 of 10 + "29%" on tap + change note "Yesterday: 1 in 10".
7. **`checkInvariants()`:** hazard, place, time window, likelihood and action present and matching the forecast → pass.
8. **Shown on S09;** the requirement traces to that screen element.

Because both days are "Possible", the word alone hides the change from 10% to 29%; this is why the profile keeps the picture and the number.

## Other research directions (for the agenda section)

A vision paper can list these as future work; each could become its own study.

| Direction | Question | Link to our work |
|---|---|---|
| Explanation under stress | How much explanation is possible once an alert is live? | Extends the alert-state layer (G124, G125) |
| Lead time | Should a 30% forecast be explained differently at 48 h, 12 h and 1 h? | New; time-series scenario |
| False alarms and trust | How to explain a flood that did not happen so the next warning is still heeded? | Appropriate reliance, Brief 1 long-term vision |
| Caregivers | One prediction, two explanations: person and helper | G80, G107 |
| Profile privacy | A saved profile may reveal a disability; who can see it? | UISE lists "privacy and ethics in adaptive UIs" |
| LLM explanation fidelity | Can generated explanations be checked automatically against the prediction and invariants? | Layer 4 of the architecture; the "accessibility illusion" |
| Language and speech | Sinhala and Tamil Easy Read explanations, read aloud | G121, G108 |

| Direction | Question | Link to our work |
|---|---|---|
| Evaluation | Requirement satisfaction, fidelity, comprehension, appropriate reliance, decision support | Brief 1 long-term vision |
| Indigenous communities | Community-defined explanation needs (source, provenance) | Brief 2, separate paper |

## Paper plan (8-page full paper)

Submission is due 13 Nov 2026 (AoE). The full-paper limit is 8 pages including references. Week-by-week work and owners are in the separate plan doc.

| Section | Pages | Content |
|---|---|---|
| Introduction | 0.75 | 30% example, problem, contributions |
| Related work | 1 | XAI for cognitive disability (NeuroAdaptX, Tielman), disaster EWS RE, our earlier paper |
| Motivating study: standard XAI vs guidelines | 1 | SHAP / LIME / counterfactual outputs, comparison table |
| Requirements and profiles | 1.25 | Task 1 triage summary, prediction-type choice, question template, three profiles |
| Framework | 1 | Framework figure, invariants, requirement → capability table |
| Prototype | 1 | Screens per profile, build method |
| Cognitive walkthrough | 1.25 | Tasks, personas, defects, agreement |
| Discussion, limitations, future work | 0.5 | No user validation yet; agenda |
| References | 0.75 | About 30 |

Other dates: notification 11 Dec 2026, camera-ready 29 Jan 2027.

## Rules to respect and open questions

### Rules from the briefs (reviewers will check these)

- Never claim profiles or explanations were validated with users; no study has been run.

- Adapt the explanation, never the prediction or its risk estimate.
- Don't assign profiles from a diagnosis.
- Mark every requirement as stated by a source or inferred by us; label hypothetical profiles as hypothetical.
- Label synthetic forecast values; don't invent evidence or data sources.
- Don't present feature importance as cause.

### Risks

- **Reviewer knows NeuroAdaptX:** without a clear contrast the paper reads as incremental. Mitigation: the positioning table and the one-line differentiation.
- **8-page limit (including references):** four tables plus two figures will not fit. Keep one summary table from Task 1 and move the full catalog triage to an online appendix.
- **One rater on the XAI comparison:** mitigated by the blind second rating on 12 Oct, with agreement reported.
- **CAP definition:** verified against OASIS CAP 1.2 §3.2.2 on 6 Oct 2026 (see task1-leftover-checks.md).

### Open questions for the team

- [ ] Does Madam agree to drop the condition picker in favour of need-based profiles, or keep both?
- [x] ~~Which river basin and forecast system for the scenario? Settled 7 Oct: Hanwella gauge, Kelani Ganga, Flood Hub–style (see Task 2).~~
- [ ] Does a second team member classify the 128 guidelines independently, so we can report agreement with Minindu's manual triage?
- [ ] Q1 (15 Oct): lead the paper with the run-time invariants and keep capability dependencies supporting, or present both equally?
- [ ] Q2 (15 Oct): alert mode keeps all five facts in one line, or only hazard, place and action (G124)?
- [ ] Q3 (15 Oct): test the AI wording check on prepared rewordings, or build a live model? Should the check require the exact CAP word?
- [ ] Where do we host the online appendix (catalog triage, profile spec)?

## Sources

**External**

- UISE 2027 call for papers

- Bunde (2026), NeuroAdaptX: Designing Neuro-Adaptive Explanations for Cognitive Accessibility in Explainable AI Interfaces, DESRIST 2026, LNCS 16605, pp. 205–225
- Tielman et al. (2024), Explainable AI for all: a roadmap for inclusive XAI for people with cognitive disabilities, Technology in Society 79, 102685
- Corrales et al. (2025), ProfileXAI: user-adaptive explainable AI
- A survey of accessible explainable AI research
- Madugalla et al. (2025), Human-Centred Requirements Engineering for Critical Systems: Insights from Disaster Early Warning Applications, arXiv:2511.12856
- Nearing et al. (2024), Global prediction of extreme floods in ungauged watersheds, Nature 627, 559–563
- Google Flood Forecasting API
- Mosavi, Ozturk & Chau (2018), Flood prediction using machine learning models: literature review, Water 10(11), 1536
- Bentivoglio et al. (2022), Deep learning methods for flood mapping, HESS 26, 4345–4378
- Irrigation Department, Hydrology and Disaster Management Division (2026, 8 Aug), Islandwide water level & rainfall situation in major rivers (DMC copy)
- OASIS (2010), Common Alerting Protocol Version 1.2, §3.2.2 (verified 6 Oct 2026)
- Method papers: Lundberg & Lee (2017) SHAP; Ribeiro et al. (2016) LIME; Wachter et al. (2017) counterfactuals; Mothilal et al. (2020) DiCE (details being checked by Minindu)


**Our materials**

- Research Brief 1 and Research Brief 2 (project docs)
- Final paper: "Designing Inclusive Disaster Early Warning Systems for People with Cognitive Impairment"
- Interim report (thesis 1)
- Competitive Analyses 01–04
- EWS design guidelines, 128 categorised
- Design compendium
- Figma prototype